Received: 1 November 2024 | Revised: 14 February 2025 | Accepted: 21 February 2025

DOI: 10.1111/dom.16311

ORIGINAL ARTICLE

WILEY

Identifying individuals at risk for weight gain using machine learning in electronic medical records from the United States

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Funding information
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Abstract

Aims: Numerous risk factors for the development of obesity have been identified, yet the aetiology is not well understood. Traditional statistical methods for analysing observational data are limited by the volume and characteristics of large datasets. Machine learning (ML) methods can analyse large datasets to extract novel insights on risk factors for obesity. This study predicted adults at risk of a ≥10% increase in index body mass index (BMI) within 12 months using ML and a large electronic medical records (EMR) database.

Materials and Methods: ML algorithms were used with EMR from Optum's de-identified Market Clarity Data, a US database. Models included extreme gradient boosting (XGBoost), random forest, simple logistic regression (no feature selection procedure) and two penalised logistic models (Elastic Net and Least Absolute Shrinkage and Selection Operator [LASSO]). Performance metrics included the area under the curve (AUC) of the receiver operating characteristic curve (used to determine the best-performing model), average precision, Brier score, accuracy, recall, positive predictive value, Youden index, F1 score, negative predictive value and specificity.

Results: The XGBoost model performed best 12 months post-index, with an AUC of 0.75. Lower baseline BMI, having any emergency room visit during the study period, no diabetes mellitus, no lipid disorders and younger age were among the top predictors for ≥10% increase in index BMI.

Conclusion: The current study demonstrates an ML approach applied to EMR to identify those at risk for weight gain over 12 months. Providers may use this risk stratification to prioritise prevention strategies or earlier obesity intervention.

KEYWORDS

BMI increase, electronic medical records, machine learning, obesity, risk factors, weight gain

1 | INTRODUCTION

The increasing prevalence of obesity is a major public health challenge associated with significant morbidity and mortality in the United States and many other countries around the world. An analysis of Global Burden of Disease (GBD) study data from over 68 million people during the period 1980 to 2015 showed that worldwide, a total of 603.7 million adults had obesity, with the prevalence doubling in over 70 countries during that period.[1] More recent GBD data also show that globally, there were an estimated 5 million deaths associated with obesity in 2019.[2] Obesity-related disability-adjusted life years (DALYs) increased each year from 2000 to 2019, with a larger increase observed in men (0.74%) than in women (0.25%).[2] Obesity-related DALYs are predicted to increase by almost 40% between 2020 and 2030.[2]

Estimates for the United States, specifically, suggest that more than one-third of the adult population has obesity[3,4] and predictions are that this proportion may increase to about half of the adult population by 2030.[5] Having obesity or pre-obesity (overweight) is associated with major chronic health complications and comorbidities, including cardiovascular disease, cancer and type 2 diabetes.[6] The underlying aetiology of obesity or pre-obesity is complex and multifactorial. Determinants for developing obesity include genetic, environmental and behavioural factors that interact and lead to excess adiposity.[3,4,6]

The forecast for a steadily increasing prevalence of obesity and pre-obesity, and the serious associated health complications, therefore creates an urgency from a public health perspective to identify groups at the highest risk for weight gain. Identifying these high-risk groups will allow healthcare systems and providers the greatest opportunity to reverse the trends of increased prevalence of obesity and pre-obesity. Large real-world datasets from electronic medical records (EMR) contain data on millions of people, including their demographics, medical history, laboratory results and medication use. The variety and availability of EMR data make them a realistic option to use to identify people who are at risk of developing or worsening disease.

Although there are statistical methods established for analysing observational data, the volume and characteristics of large datasets from real-world sources (e.g., the unevenness of data completeness) have increased the need for the utility of new analytical approaches such as machine learning (ML).[7] ML uses a family of mathematical and statistical methods to characterise and analyse large amounts of data, control for confounders, identify and learn from patterns, and thereby generate precise predictions of risk factors.[3,7–11] ML can complement traditional statistical methods to improve prediction accuracy and is superior to traditional regression methods in several ways, including the ability to handle complex and non-linear relationships, high-dimensional data and missing data.[7,8,12]

ML is also becoming increasingly accessible and easier to use, and there have been applications of ML in the obesity research field using datasets from different countries[8,13–20]; however, the extent to which these data are applicable to the general US population is uncertain. Therefore, the objective of this study was to predict adults at risk of weight gain in the short term, defined as a ≥10% increase in their index BMI within 12 months, using a large EMR database from the United States.

2 | MATERIALS AND METHODS

2.1 | Study design

2.1.1 | Data source

This study used EMR data from Optum's de-identified Market Clarity Data (Market Clarity; a fully Health Insurance Portability and Accountability Act compliant, statistician-certified and de-identified dataset). These EMR data are nationally representative of the United States and are sourced from ambulatory and Integrated Delivery Networks provider systems. The available information includes medications prescribed and administered, laboratory results, vital signs, body measurements, diagnoses, procedures and information from physician notes using Natural Language Processing.

2.1.2 | Study population

The study population was extracted from three non-mutually exclusive subgroups in the Market Clarity dataset: people with obesity, people with pre-obesity and a normal BMI group. The obesity subgroup was made up of anyone with at least one BMI record ≥30 kg/m^2 and the pre-obesity subgroup included anyone with at least one BMI record 27–29.9 kg/m^2 to align with definitions of pre-obesity as excessive accumulation of body fat that can impair health[21] and as per a guideline-based approach to prescribing approved anti-obesity medications.[22] The normal BMI subgroup was made up of a 5% random sample of those with at least one BMI measurement between ≥18.5 and <25 kg/m^2.

The study design is summarised in Figure 1A. People with at least two BMI measurements of ≥18.5 to <40 kg/m^2 recorded between 3 months and 2 years apart, and stable body weight (defined as absolute difference ≤3% between the BMI measurements) were identified. This is consistent with previous studies which have defined stable weight as ≤3% change over time.[23,24] The study period was from 1 January 2010 to 31 December 2020. The index date was set on the second BMI measured, and baseline was defined as 12 months pre-index. A third BMI measurement (follow-up BMI) was at least 3 months post-index (second BMI measurement). Eligible people were aged 19–65 years at index, and all participants had EMR activity prior to 12 months pre-index and after 12 months post-index.

Individuals with a BMI of ≥40 kg/m^2 (class III obesity) during the baseline period were not included in the study because this is a complex condition with multiple contributing factors and potentially includes confounding features, which make it difficult to determine the true effect of features of interest. Other exclusion criteria were people with missing sex data and those with a pregnancy or delivery diagnosis or procedure code during the 12 months pre- through 12 months post-index or a COVID diagnosis code during the 12 months pre- through 12 months post-index.

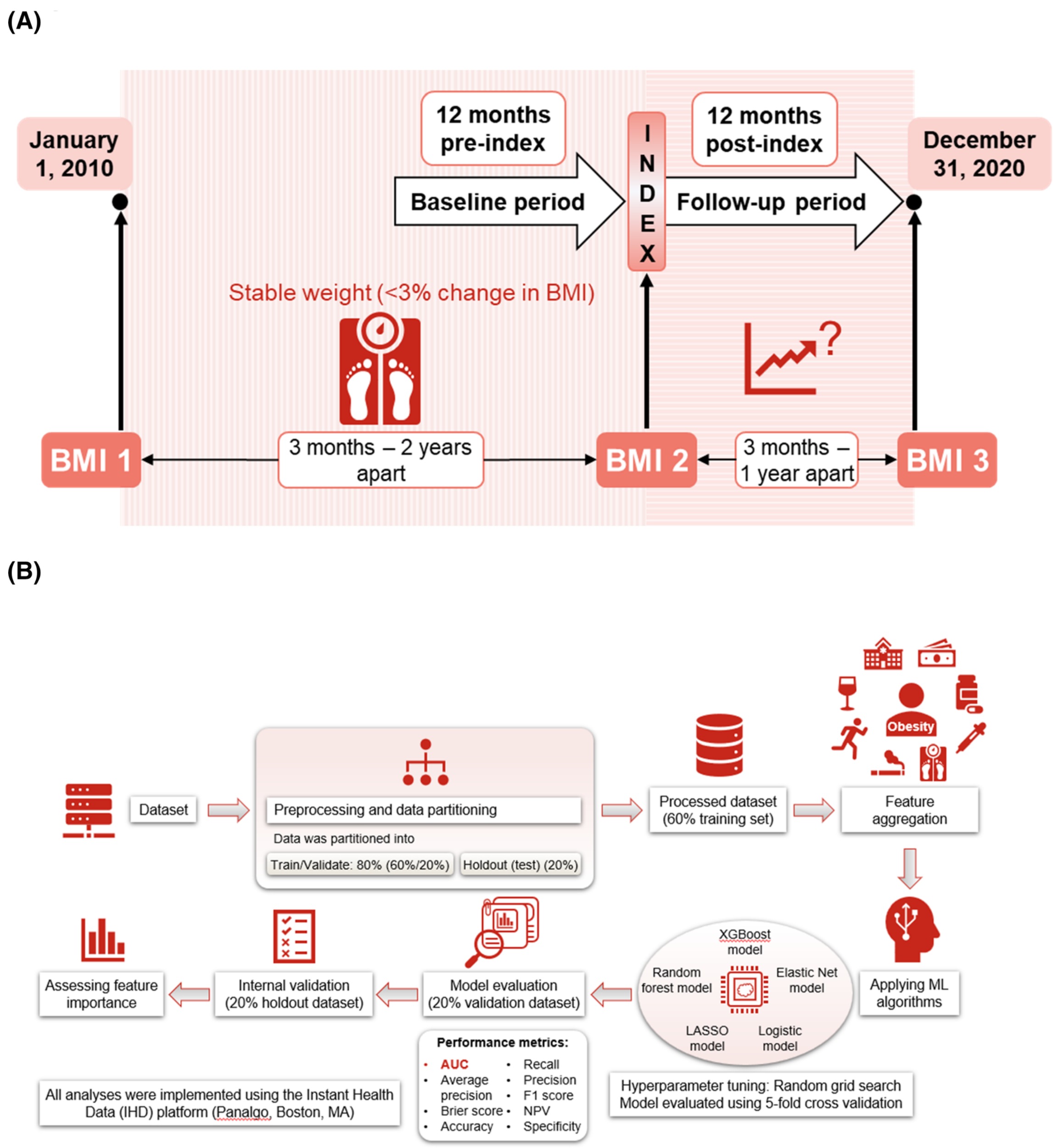

FIGURE 1 (A) Study design. (B) Schematic of ML methods used. Missing values for continuous predictors were excluded. For categorical type measures, the "missing" category was added. AUC, area under curve; BMI, body mass index; LASSO, Least Absolute Shrinkage and Selection Operator; ML, machine learning; NPV, negative predictive value; XGBoost, extreme gradient boosting.

2.1.3 | Outcomes of interest

Weight gain was based on percentage change in BMI at the individual level. Percentage change in BMI over time for individual people was followed as a corollary to percentage change in body fat. Although BMI is not an exact measurement of body fat and has limitations, it was chosen because this data were more prevalent in the dataset used. The primary outcome was a ≥10% increase in index BMI during the 12-month post-index period. A ≥10% increase was chosen as the primary outcome because a >10% body weight change is associated with clinically meaningful outcomes such as risks for cardiovascular events.[25] Weight gains during the post-index period were also stratified by percentage BMI increase (>3% and <5% from index BMI, ≥5% and <10% from index BMI and ≥10% from index BMI), with ML methods applied to predict the factors associated with each category change.

2.1.4 | Model features

Features included in the modelling were: BMI measurements (initial and index BMI, post-index BMI and the stratified categories of weight gain from index BMI), demographics at index (age, sex, index year, race, ethnicity, payors, region and census division), clinical characteristics during baseline (diagnoses [i.e., clinical classification software], medications [i.e., Multum], procedures [i.e., healthcare common procedure coding system], current procedure terminology, lifestyle factors [e.g., alcohol intake and physical activity], laboratory tests and results and natural language processing from unstructured clinical notes) and healthcare resource utilisation during baseline (number of inpatient visits, number of outpatient visits, number of emergency room visits and number of specialist visits).

2.2 | ML methods

An overview of ML methods is shown in Figure 1B.

2.2.1 | Analytical tool

All analyses were implemented using the Instant Health Data platform (Panalgo, Boston, MA, USA). Several families of algorithms/models were used in the ML framework: extreme gradient boosting, random forest, simple logistic regression (no feature selection procedure) and two penalised logistic regression models (Elastic Net and Least Absolute Shrinkage and Selection Operator).

2.2.2 | Feature aggregation

To increase the ease of computation and clinical interpretation, features drawn from EMR datasets were combined into relevant code hierarchies or clinical concepts using clinically meaningful categories in ICD-9 or ICD-10-CM defined by Clinical Classifications Software Refined categories, and subcategories by the Healthcare Cost and Utilization Project, a Federal-State-Industry partnership sponsored by the Agency for Healthcare Research and Quality.[26]

2.2.3 | Modelling

Data were partitioned into train/validate (60%/20%) and holdout (test) sets (20%) (Figure 1B). Partitioning of data allows learning of patterns from the data, provides insights on how to tune the model and, ultimately, understand how the model would perform in newly introduced unseen data.[7,27] Hyperparameters are values used to control the learning process that are tunable and can directly affect how well a model was trained. Hyperparameter tuning used a random grid search to find the best settings for the ML models by exploring a defined range of values for each hyperparameter and evaluating the model's performance (area under the curve [AUC]) on the validation dataset. Details of the hyperparameters used are shown in Table S1. Five-fold cross-validation was used to tune hyperparameters for each ML model, then the performance for different models was evaluated on the single validation 20% dataset. Final validation of the best-performing model was conducted using the test dataset (20%), which was not used during the model's training and validation. This approach provided a good proxy for how the model would perform with new data.

2.2.4 | Model evaluation

Suitable performance metrics, as previously used in published studies of ML in the prediction of obesity[10,11,17–19,28] were used for the current study. AUC of the receiver operating characteristics curve was used to determine the best-performing model. Additionally, average precision, Brier score, accuracy, recall, precision (positive predictive value), Youden index, F1 score, negative predictive value and specificity were included. No data were imputed.

2.2.5 | Statistical methods

We present the descriptive statistics stratified by change in BMI (</≥10%). Continuous features were summarised as mean, standard deviation and median. We used the t-test to compare continuous features (i.e., age at index) and Chi-square tests to compare categorical features between BMI (</≥10%) groups. Missing values were excluded for continuous measures and included in a separate category "Missing" for categorical measures.

3 | RESULTS

Participant inclusion criteria and attrition are shown in Figure 2. A 50% sample was chosen in the final analysis to reduce processing time and resource requirements since ML models are effective at identifying common patterns in a smaller sample. From the population of 26,680,059 people in the EMR database, 926,791 were included in the study cohort. Of these, 2.9% had ≥10% BMI increase from the index BMI during the 12-month post-index period; 76.8% had <3% BMI increase; 11.1% had a 3 to <5% BMI increase and 9.1% had a 5 to <10% BMI increase. Histograms depicting the distribution of baseline BMI, stratified by BMI change (</≥10%), for the study population overall and by sex and age group (</≥50 years) are presented in Figure S1.

The characteristics of people in the study cohort overall and according to the change in BMI during follow-up (<10% and ≥10% increase from the index BMI during the 12-month post-index period) are shown in Table 1. Most of the cohort were in the pre-obesity/obesity spectrum. All characteristics considered were statistically significantly different between the two weight-change cohorts (p < 0.0001), and people with ≥10% increase in BMI were younger and more likely to be female. In the overall cohort, 5.3% had index BMI values in the normal range, and 94.7% were classified as having pre-obesity or obesity based on index BMI. Among the overall population, 78.7% had outpatient visits, 7.9% had inpatient visits and 17.1% had emergency room visits during the baseline period. The

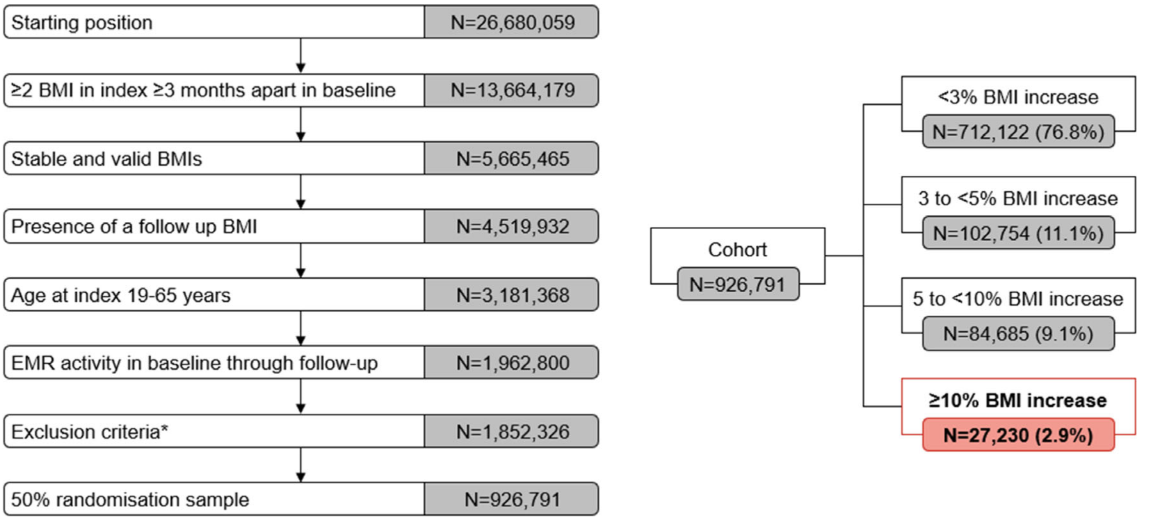

FIGURE 2 Participant inclusion criteria and starting population. *Missing sex ($n = 693$) or a pregnancy ($n = 108\,874$) or a COVID-19-related medical encounter ($n = 907$) in pre- through post-index. BMI, body mass index; COVID-19, coronavirus disease 2019; EMR, electronic medical record.

characteristics of people in the study cohort according to weight gain <3%, ≥3%; <5%, ≥5%; and <10%, ≥10% from index BMI are shown in Table S2.

3.1 | Model metrics

The XGBoost model performed the best of the five ML models for predicting ≥10% increase in the index BMI at the 12-month post-index follow-up, with an AUC of 0.75 (Table 2). Higher AUC represents better distinction between people who had ≥10% BMI increase and <10% increase. A lower Brier score represents greater accuracy. At the 12-month post-index follow-up, the XGBoost model demonstrated moderate predictive ability in cohorts with >3% and >5% BMI increase, with AUCs of 0.62 and 0.66, respectively (Table S3). Feature importance for the top 20 features identified by the XGBoost model, reflecting the strength of the absolute association between each feature and a ≥10% BMI increase from the index BMI during the 12-month post-index period, measured by Gain, is shown in Figure 3.

The most important risk factors for a ≥10% increase in BMI were index BMI, followed by the presence of emergency room visits, a diagnosis of diabetes and lipid disorders and age at index (Figure 3). Specifically, those with a higher index BMI value, a diagnosis of diabetes and lipid disorders and an increased age at index had a lower chance of a ≥10% increase in BMI. Conversely, the chance of this BMI increase was higher when emergency room visits were present (Figure 3).

In addition, having inpatient visits, being female, being a Medicaid recipient, having a concomitant substance-related disorder diagnosis, undergoing an emergency room procedure, having a concomitant mood disorder diagnosis or having a concomitant CNS disorder diagnosis were identified as other important risk factors that increased the chances of a ≥10% BMI increase. Conversely, important risk factors that reduced the risk of a ≥10% increase from the index BMI during the 12-month post-index period were having at least one diagnostic test (most commonly prescriptions for glucose test strips), annual wellness check/preventive visit, lipid profile test, Asian race, being commercially insured and receipt of a prescription for antihyperglycaemic, antihyperlipidaemic or progestin agent.

4 | DISCUSSION

The current study demonstrates that using ML methods with real-world data can identify people at risk for weight gain (defined as ≥10% increase in their index BMI) within 12 months. This confirmed that ML is a powerful tool to identify the best fitting model and extract insights from large amounts of data. Our study used EMR data, a readily available source of clinical information, from a nationally representative US dataset (Market Clarity), although there is the potential to use the same approach with other datasets.

BMI was used as a surrogate for body fat.[29,30] because BMI data were readily available through the EMR, whereas measured body fat percentage is infrequently available. By following BMI data on an individual basis among adults, where it is assumed height remains constant, a 10% increase in BMI can be correlated with a 10% change in weight.

At the index date, the majority of the people in the study cohort (>94%) could be characterised as having excess adiposity based on BMI. Overall, 23.2% of the study cohort had an increase in BMI of ≥3% and 2.9% had an increase in BMI of ≥10% in 12 months. Baseline BMI significantly differed between those with versus without ≥10% increase in BMI ($p < 0.0001$). Among people with ≥10% increase in BMI, baseline BMI was in the normal range for 15.8% and in the pre-obesity range for 41.6% (as compared with baseline BMI in the normal range for 5.0% and in the pre-obesity range for 27.7%, respectively, among those in the group with <10% BMI increase).

TABLE 1 Population characteristics overall and by change in BMI (</≥10%).

	<10% BMI change in follow-up	≥10% BMI change in follow-up	Overall	*p*-value
Cohort, *N* (%)	899 561	27 230	926 791	<0.0001
Age at index				<0.0001
Mean (SD)	48.1 (11.8)	42.9 (13.3)	47.9 (11.9)	
Median	50	44	50	
Sex, *n* (%)				<0.0001
Female	474 600 (52.8)	17 096 (62.8)	491 696 (53.1)	
Race, *n* (%)				<0.0001
Asian	14 287 (1.6)	272 (1.0)	14 559 (1.6)	
Black	109 944 (12.2)	4132 (15.2)	114 076 (12.3)	
Caucasian	712 336 (79.2)	20 889 (76.7)	733 225 (79.1)	
Missing	62 994 (7)	1937 (7.1)	64 931 (7.0)	
Ethnicity, *n* (%)				<0.0001
Hispanic	50 489 (5.6)	1811 (6.7)	52 300 (5.6)	
Not Hispanic	783 019 (87.0)	23 644 (86.8)	806 663 (87.0)	
Missing	66 053 (7.3)	1775 (6.5)	67 828 (7.3)	
Region, *n* (%)				<0.0001
Midwest	479 198 (53.3)	14 651 (53.8)	493 849 (53.3)	
Northeast	111 697 (12.4)	3163 (11.6)	114 860 (12.4)	
South	194 154 (21.6)	5647 (20.7)	199 801 (21.6)	
West	78 572 (8.7)	2264 (8.3)	80 836 (8.7)	
Missing	35 940 (4.0)	1505 (5.5)	37 445 (4.0)	
Payor, *n* (%)				<0.0001
Commercial	510 032 (56.7)	12 564 (46.1)	522 596 (56.4)	
Medicaid	56 790 (6.3)	3203 (11.8)	59 993 (6.5)	
Medicare	49 577 (5.5)	1664 (6.1)	51 241 (5.5)	
Uninsured	37 684 (4.2)	1717 (6.3)	39 401 (4.3)	
Unknown	245 478 (27.3)	8082 (29.7)	253 560 (27.4)	
Smoking status, *n* (%)				<0.0001
Current smoker	110 949 (12.3)	4787 (17.6)	115 736 (12.5)	
Never smoked	271 902 (30.2)	7266 (26.7)	279 168 (30.1)	
Previous smoker	163 878 (18.2)	4034 (14.8)	167 912 (18.1)	
Unknown	352 832 (39.2)	11 143 (40.9)	363 975 (39.3)	
Index BMI value				<0.0001
Mean (SD)	31.8 (4.0)	29.4 (4.5)	31.7 (4.0)	
Median	31.6	29.1	31.5	
Index BMI group				<0.0001
Normal (BMI 18.5–25 kg/m^2)	44 726 (5.0)	4292 (15.8)	49 018 (5.3)	
Pre-obesity (BMI 27–30 kg/m^2)	249 087 (27.7)	11 325 (41.6)	260 412 (28.1)	
Obesity (BMI 30–40 kg/m^2)	605 748 (67.3)	11 613 (42.6)	617 361 (66.7)	
Outpatient visits in baseline, *n* (%)	709 825 (78.9%)	19 559 (71.8%)	729 384 (78.7%)	<0.0001
Number of outpatient visits				0.0036
Mean (SD)	5.4 (8.4)	5.6 (9.8)	5.4 (8.4)	
Median	3	2	3	
Inpatient visits in baseline, *n* (%)	69 394 (7.7%)	3555 (13.1%)	72 949 (7.9%)	<0.0001
ER visits in baseline, *n* (%)	151 104 (16.8%)	7833 (28.8%)	158 937 (17.1%)	<0.0001

Note: T-test was used to compare age at index; Chi-square test was used for comparison of categorical features.
Abbreviations: BMI, body mass index; ER, emergency room; SD, standard deviation.

TABLE 2 Performance of predictive models for identifying individuals at risk of a ≥10% increase in the index BMI during the 12-month post-index period.

Model type	# of features	AUC (95% CI)	Average precision	Brier score	Accuracy	Recall	Precision (PPV)	Youden index	F1 score	NPV	Specificity
Performance of predictive models fitted to the training set (60%) and evaluated on the validation dataset (20%)											
XGBoost model	349	0.74 (0.73, 0.75)	0.09	0.0275	0.72	0.64	0.07	0.36	0.12	0.99	0.72
Elastic net model	286	0.73 (0.72, 0.73)	0.08	0.0277	0.68	0.66	0.06	0.34	0.11	0.99	0.68
Logistic model	354	0.73 (0.72, 0.73)	0.08	0.0277	0.69	0.64	0.06	0.34	0.11	0.98	0.69
Random forest model	353	0.73 (0.72, 0.74)	0.08	0.0278	0.68	0.66	0.06	0.34	0.11	0.99	0.68
LASSO model	227	0.73 (0.72, 0.73)	0.09	0.0277	0.69	0.65	0.06	0.34	0.11	0.98	0.69
Performance of the predictive models fitted to the test dataset (20%)											
Model type	**# of features**	**AUC (95% CI)**	**Average precision**	**Brier score**	**Accuracy**	**Recall**	**Precision (PPV)**	**Youden index**	**F1 score**	**NPV**	**Specificity**
XGBoost model	349	0.75 (0.74, 0.75)	0.10	0.0278	0.72	0.65	0.07	0.37	0.12	0.98	0.72
Elastic net model	286	0.73 (0.72, 0.74)	0.09	0.0281	0.68	0.67	0.06	0.35	0.11	0.99	0.68
Logistic model	354	0.73 (0.72, 0.74)	0.09	0.0281	0.69	0.65	0.06	0.34	0.11	0.98	0.69
Random forest model	353	0.73 (0.73, 0.74)	0.09	0.0282	0.68	0.67	0.06	0.35	0.11	0.99	0.68
LASSO model	227	0.73 (0.72, 0.74)	0.09	0.0281	0.69	0.65	0.06	0.35	0.11	0.98	0.69

Note: AUC of the receiver operating characteristics curve measures discrimination (higher AUC = better distinction between patients with and without obesity). Average precision summarises the precision-recall plot as a weighted mean of precisions achieved at each threshold. Feasible values are between 0 and 1 with higher values indicating better performance. Brier score is the predicted probability outputted by a model that matches the observed probability (lower Brier score = greater accuracy). Accuracy is the binary classification accuracy (i.e., ratio of the number of correctly classified predictions over the number of total samples). Higher recall = better maximisation of the number of true positives. Precision (PPV) is the proportion of positive observation (true positive) the model correctly identified from all the observations it labelled as positive (higher precision = better minimisation of false positives). Youden index represents a summary measurement of the receiver operating characteristic curve for the accuracy of a test with ordinal or continuous endpoints and gives equal weight to false-positive and false-negative values, so when two tests have the same index, it means they have the same proportion of total misclassified results. Higher F1 score = better performance of model. NPV is the probability of a true negative. Specificity is the probability of a true negative (higher specificity = better identification of negative results).
Abbreviations: AUC, area under curve; CI, confidence interval; LASSO, Least Absolute Shrinkage and Selection Operator; NPV, negative predictive value; PPV, positive predictive value; XGBoost, extreme gradient boosting.

Our study found that having a lower baseline BMI, having any emergency room visit during the study, no diabetes mellitus, no lipid disorders and a younger age were among the most important factors associated with a gain of ≥10% of index BMI. These risk factors, overall, could suggest those early in the obesity disease spectrum. Furthermore, the population with these risk factors may not have regular interaction with healthcare providers or the healthcare system and, therefore, may be more susceptible to disease progression. Other important risk factors that were found to be associated with BMI increase were having an inpatient visit, being female, undergoing an emergency room procedure, being a Medicaid recipient, and having a substance-related disorder, mood disorder or CNS disorder diagnosis. These risk factors likely emphasise the multifactorial pathogenesis of obesity, which is influenced by a variety of considerations, including psychosocial factors, social determinants of health and the contribution of comorbid conditions and their management. However, these observations require further investigation, and it will be important to explore how to use the risk factors identified in our study to improve the identification of those at risk for significant weight gain.

By identifying the risk factors for increases in BMI and introducing strategies to mitigate these factors, we may be able to slow the rising prevalence of obesity and improve the health of our population. For example, this study identified that younger adults with a lower baseline BMI were at increased risk of gaining ≥10% of their index BMI. In adults, a ≥10% increase in BMI over the period of 12 months is significantly greater than expected age-related increases in body

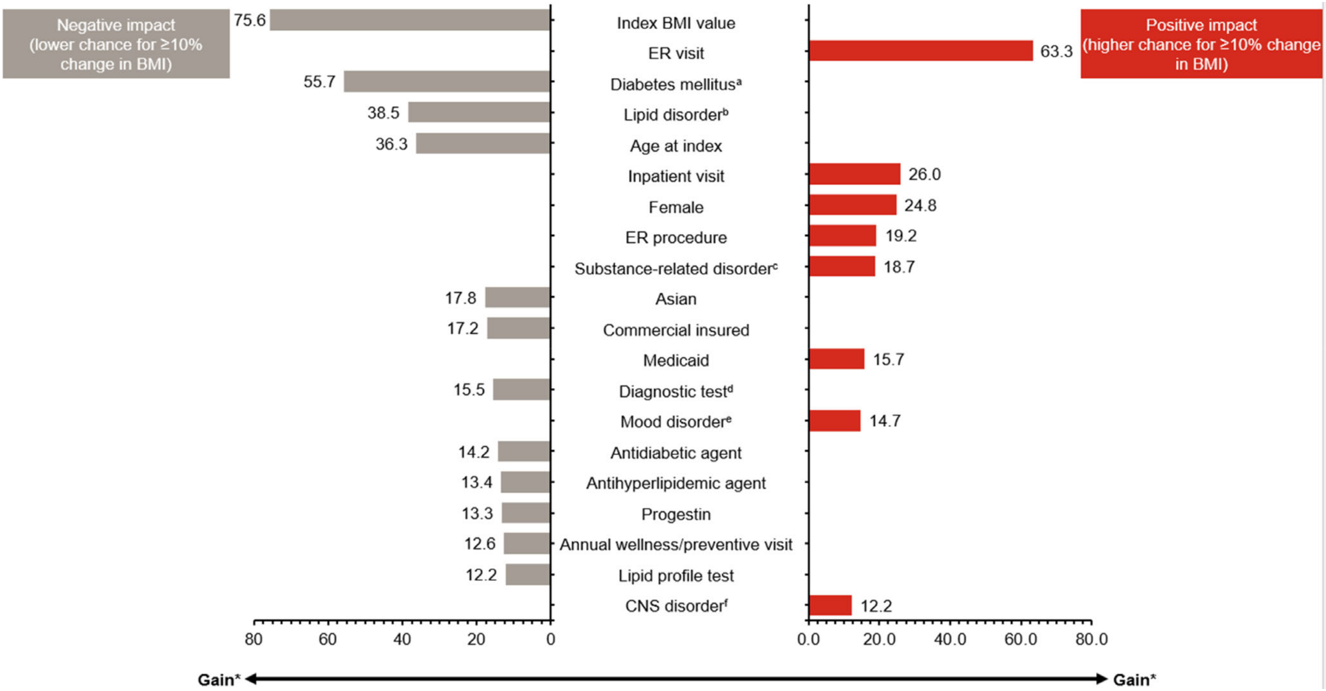

FIGURE 3 Top 20 features of importance identified by XGBoost from the validation dataset. *Gain is the relative contribution of the corresponding feature to the model. Specifically, gain is the improvement in prediction accuracy achieved by adding a particular feature to a model compared to using a simpler model without that feature, and it tells how much better the model predicts with the additional feature. All features were captured in baseline, except when indicated as index. Gain is calculated as the difference in the loss function before and after a split on a feature. The higher the Gain, the more important the feature is to the model. In other words, Gain is the improvement in accuracy brought by a feature to the branches. [a]Most commonly T1DM, T2DM and other diabetes mellitus without complications. [b]Most commonly hyperlipidaemia and hypercholesterolaemia. [c]Most commonly tobacco use disorder, alcohol abuse and drug abuse. [d]Most commonly prescriptions for glucose test strips. [e]Most commonly major depressive disorder. [f]Most commonly chronic pain and migraine. This illustrates the most significant features in descending order, with the features at the top of the figure contributing more to the model and having high predictive power for a ≥10% BMI increase from the index BMI during the 12-month post-index period compared to the features shown lower in the figure. BMI, body mass index; CNS, central nervous system; ER, emergency room; T1DM, type 1 diabetes mellitus; T2DM, type 2 diabetes mellitus; XGBoost, extreme gradient boosting.

weight.[31] This suggests that we may need to focus on primary prevention efforts in this population to prevent excessive weight gain and ensure that weight management interventions are accessible to underserved populations, including those who are Medicaid recipients, those from ethnic minorities and people with mental health conditions.[32–35] Given the XGBoost model specificity of 72%, obesity prevention could range from a soft approach involving education and monitoring to active interventions, depending on the characteristics and behaviours of the individual concerned.

The suggestion that a focus on the prevention of obesity should include attention to younger individuals has been supported by other investigators who examined factors linked to weight gain.[14,36] One of these studies used ML and linked EMR data obtained from the Clinical Practice Research Datalink in England to identify populations at risk for BMI change over 1-, 5- and 10-year increments. These authors reported that the factor most associated with an increase in BMI was young-adult age (18–24 years); sex and race were also significantly associated with the risk of gaining weight.[14] However, this study included only a few potential features for assessing the risk of weight gain, including age, sex, race/ethnicity, region and degree of social deprivation. Additionally, the study population was from a British database, so the results might not be generalisable to US populations. Williamson et al.[36] used data from the First National Health and Nutrition Examination Survey Epidemiologic Follow-up Study to estimate the 10-year incidence of major weight gain and found the occurrence to be greatest in females, particularly those who were overweight and those aged 25 to 34 years. These findings led the investigators to conclude more than 30 years ago that obesity prevention should begin among adults in their early 20s and that special emphasis is needed for young women with overweight.[36]

The characteristics of people included in our study were broadly comparable to other managed care populations in the United States. For example, a recent retrospective, descriptive analysis reported characteristics and weight-change patterns of a managed care population classified with pre-obesity or obesity at a large IDN (Baylor Scott and White Health) in the state of Texas.[23] In a population of 22,712 people who were continuously enrolled for the entire period of that study, the mean age at the index date was 57.2 years, 55.4% were female and over half were commercially insured. Common comorbidities among the study cohort included type 2 diabetes, dyslipidaemia, cardiovascular disease, hypertension and psychiatric disorders. Mean BMI increased from 32.8 kg/m^2 at the index date to 33.4 kg/m^2 after 12 months, and 23.7% of people gained more than 3% of body weight, similar to the 23.2% in our study who had an increase in BMI of ≥3%. However, the population in the study from Osterland et al.[23] had a low reported frequency of ER (0.11%) or inpatient visits (1.68%) (defined as at least one medical claim for an ER or inpatient visit with 12 months of index date) when compared to the cohort in our study (~17% of the population in our study had ER visits and ~8% inpatient visits in the study period).

The aetiology of obesity is heterogeneous and complex, with complications and metabolic manifestations varying among individual people.[3,6,37] The relationship between features may not be linear, which complicates the prediction of risk factors for obesity based on observational data.[37] In this context, ML methods are especially suited modelling tools to find complex, non-linear relationships between the predictor and response features and make predictions about individuals at risk for obesity because of the complex relationships between many features.[12]

4.1 | Limitations

The quality and quantity of the data (including patient characteristics and behaviours) used for training could significantly impact the model's performance. In addition, the model's performance may not be sufficient for all applications and is best suited as a decision support tool to assist physicians or payers in identifying patients at risk for weight gain. Several factors can influence the model's predictivity, including: (1) data limitations such as imbalanced class distributions and missing data, which were addressed using techniques such as oversampling and cross-validation to mitigate overfitting and improve reliability; (2) the inclusion of numerous features and the careful application of feature engineering to enhance interpretability and predictive power; and (3) data heterogeneity arising from variations in patient populations, clinical practices and data collection methods, which can degrade performance across different settings. However, the EMR data used in this study was well curated, which has the potential to improve the model's accuracy, even with a moderate AUC.

The dataset for the study defined the pre-obesity subgroup to include anyone with at least one BMI record 27 to 29.9 kg/m^2. This aligns with definitions of pre-obesity as excessive accumulation of body fat that can impair health[21] as per a guideline-based approach to prescribing approved obesity medications.[22] However, this cut-off skewed the starting population towards the individuals with pre-obesity/obesity and means that the population included are not representative or generalisable for the US population. The findings of this study are also subject to limitations pertaining to usage of real-world data and to missing data on key features that could potentially predict weight gain (e.g., diet, physical activity). It has been suggested that predictive ability may be improved by use of multiple ML algorithms.[10] Moreover, the results of this study cannot be generalised to individuals outside the United States or to those without a healthcare encounter. This study also required that the people included have at least three BMI measurements, which is a potential source of selection bias as it includes only people with a more frequent level of healthcare interaction.

Because of these model limitations, it is crucial to emphasise the importance of human judgement in interpreting and acting upon the model's predictions, ensuring that it serves as a valuable aid rather than an autonomous decision-maker.

5 | CONCLUSIONS

Obesity is a complex disease with multiple contributing factors that lead to significant morbidity and mortality. Utilising strategies that focus on those populations most likely to gain excess weight can help better utilise public health and healthcare resources for obesity prevention. Individual characteristics that can predict weight-gain risk, such as clinical history, certain demographics and comorbid conditions may be readily available through EMR. The current study demonstrates an ML approach applied to EMR that can be used to identify the characteristics of individuals at risk of an increasing BMI and provide opportunities for obesity prevention. Ideally, we would test the model's performance on different datasets to assess its robustness and generalisability. Nevertheless, researchers or large hospital networks could potentially apply similar approaches to develop a model to profile their own patients at risk. We identified lower baseline BMI, having any emergency room visit during the study, no diabetes mellitus, no lipid disorders and younger age as among the most important factors associated with a gain of ≥10% of index BMI. Consequently, our findings suggest that it may be prudent that the prevention of obesity include attention to younger individuals with relatively low weight (BMI).

AUTHOR CONTRIBUTIONS

Casey Choong was involved with the conception and design of the work, and the acquisition and interpretation of the data. Neena Xavier and Beverly Falcon were involved with the interpretation of the data for the work. Hong Kan was involved with the conception of the work and the interpretation of the data. Ilya Lipkovich was involved with the design of the work and the interpretation of the data. Callie Nowak, Christy Houle and Scott Kahan were involved with the analysis and interpretation of the data for the work. Margaret Hoyt was involved with the design of the work and the analysis and interpretation of the data. All authors provided critical revision of the manuscript for important intellectual content and have participated sufficiently in the work to agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. All authors have given their final approval of the manuscript to be published.

ACKNOWLEDGEMENTS

The authors would like to acknowledge Mark O'Connor and Caroline Spencer (Rx Communications, Mould, UK) for medical writing assistance with the preparation of this manuscript.

FUNDING INFORMATION

This work was funded by Eli Lilly and Company.

CONFLICT OF INTEREST STATEMENT

Casey Choong, Neena Xavier, Beverly Falcon, Hong Kan, Ilya Lipkovich, Callie Nowak, Margaret Hoyt and Christy Houle are full-time employees and stockholders of Eli Lilly and Company. Scott Kahan has acted as a consultant for Carmot, CorEvitas, Currax, Eli Lilly and Company, Vivus and WW.

PEER REVIEW

The peer review history for this article is available at https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/dom.16311.

DATA AVAILABILITY STATEMENT

The datasets generated and/or analysed during the current study are not publicly available due to individual data privacy but may be available from the corresponding author on reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Choong C, Xavier N, Falcon B, et al. Identifying individuals at risk for weight gain using machine learning in electronic medical records from the United States. *Diabetes Obes Metab*. 2025;27(6):3061-3071. doi:10.1111/dom.16311